Internet of Things Home Security System

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CEIS101 Final Course Project

Fall 2023

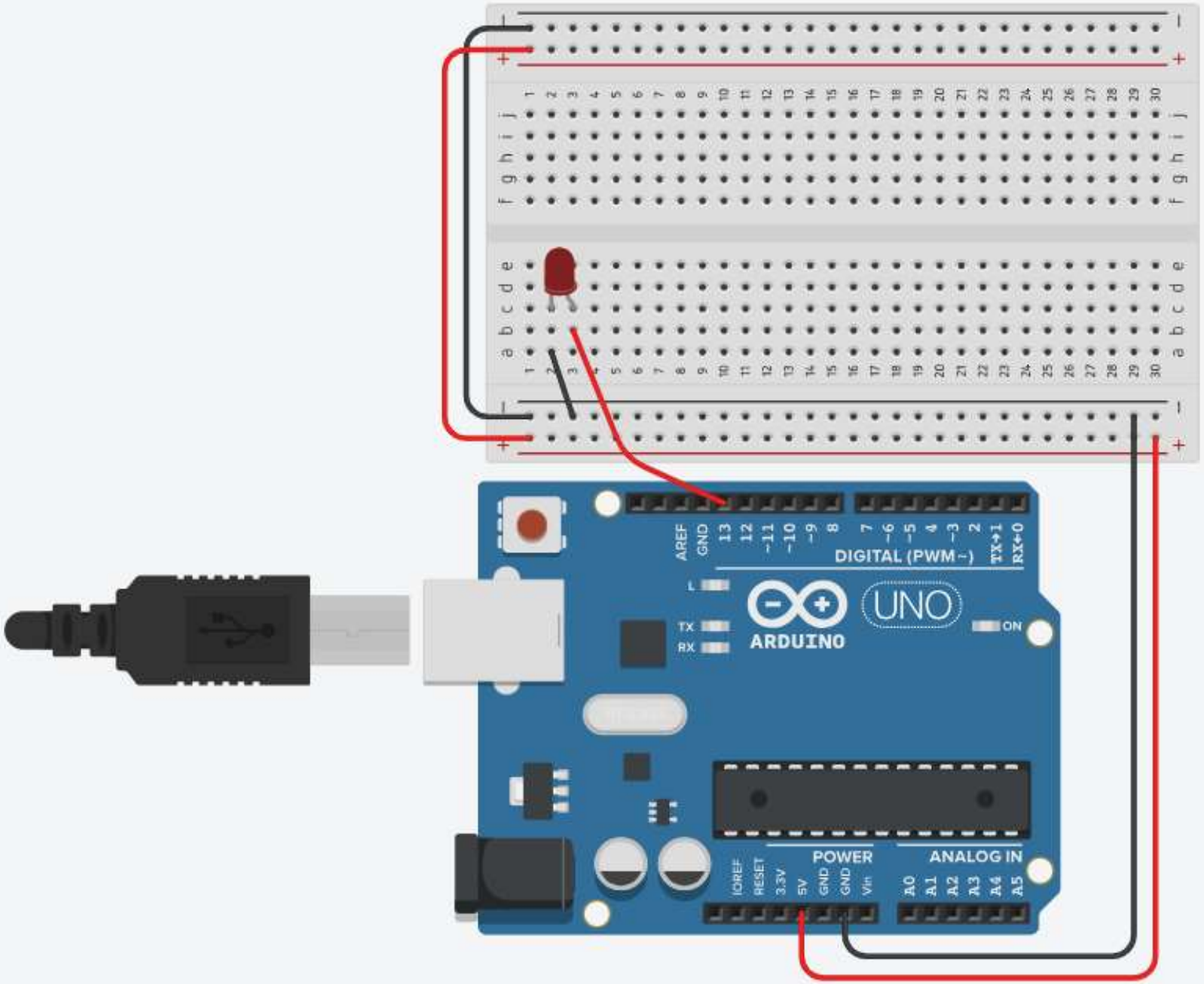
Prof. Vijay Deokar

Introduction

The objective of this project was to mimic an IoT home security system utilizing an Arduino Mega 2560 and its corresponding IDE in conjunction with the C programming language to assign physical hardware particular responsibilities.

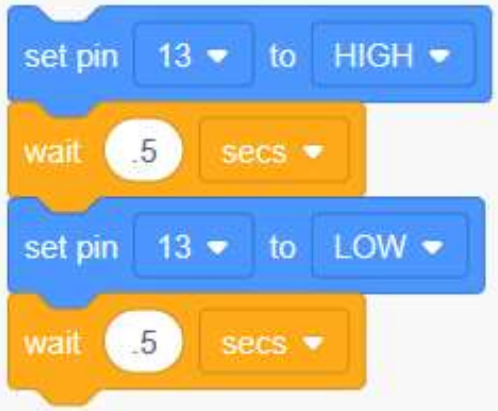
Tinkercad

The initial phase was the utilization of Tinkercad to familiarize myself with building circuitry.



Tinkercad

Code and code blocks created for circuit testing.

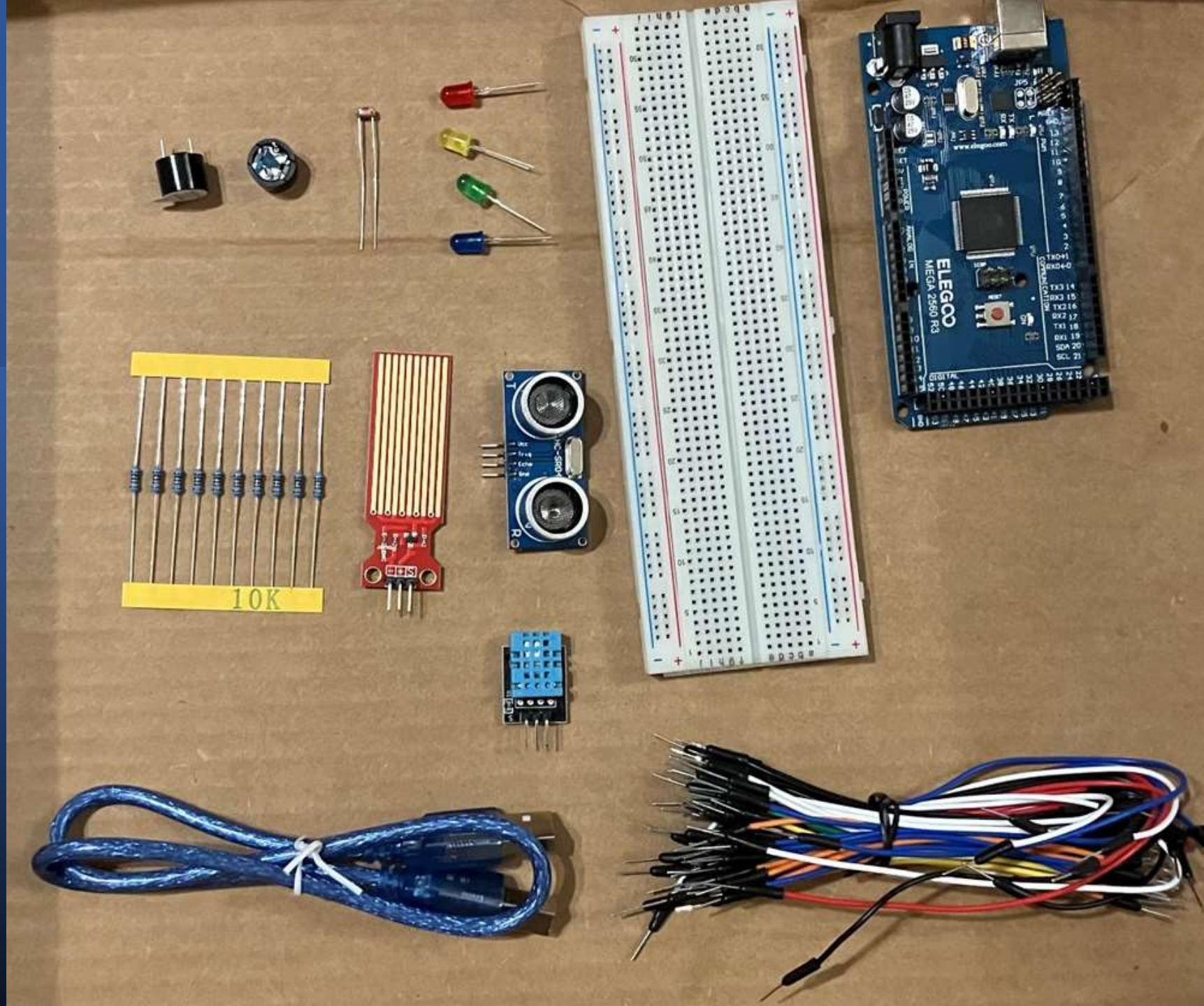



```
// C++ code
//
void setup()
{
  pinMode(13, OUTPUT);
}

void loop()
{
  digitalWrite(13, HIGH);
  delay(500); // Wait for 500 millisecond(s)
  digitalWrite(13, LOW);
  delay(500); // Wait for 500 millisecond(s)
}
```

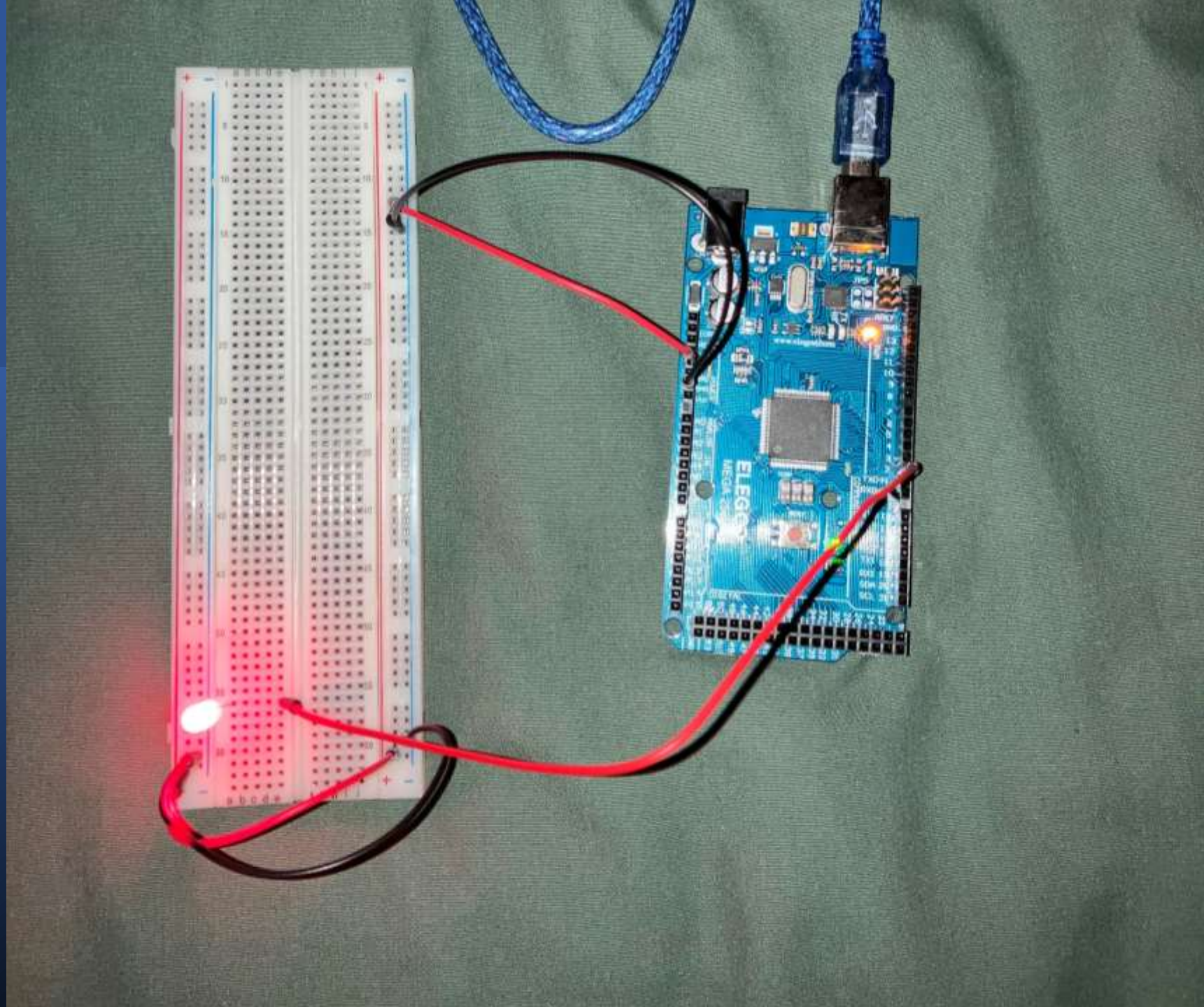
Organization

Hardware included an Arduino Mega 2560, a breadboard, 10kΩ resistors, LEDs, an ultrasonic sensor, buzzers, wires, a photoresistor, a USB Type B cable, a DHT-11 sensor and a water level sensor.



Red LED

A red LED light was programmed to illuminate.



Red LED Light Serial Monitor

Arduino IDE output of the red LED light.

LEDSerial.ino

```
1   #define Rled 2 // Set up variable for Red LED as pin 2
2
3   void setup() {
4   Serial.begin(9600); // Sets the baud rate for logging
5   pinMode(Rled, OUTPUT); //Set up Rled (pin 2) as output
6   Serial.println("CEIS101 Course Project Module 3");
7   Serial.println("Name: Francisco Acosta "); //replace xxxxx with your name
8   }
9
10  void loop() {
11  digitalWrite(Rled, HIGH); // Turn ON the Red LED
12  Serial.println("Red LED is ON");
13  delay(1000);//delay 1000 msec= 1 second
14  digitalWrite(Rled, LOW);  // Turn OFF the Red LED
15  Serial.println("Red LED is OFF");
16  delay(1000); //delay 1000 msec= 1 second
17  }
18
```

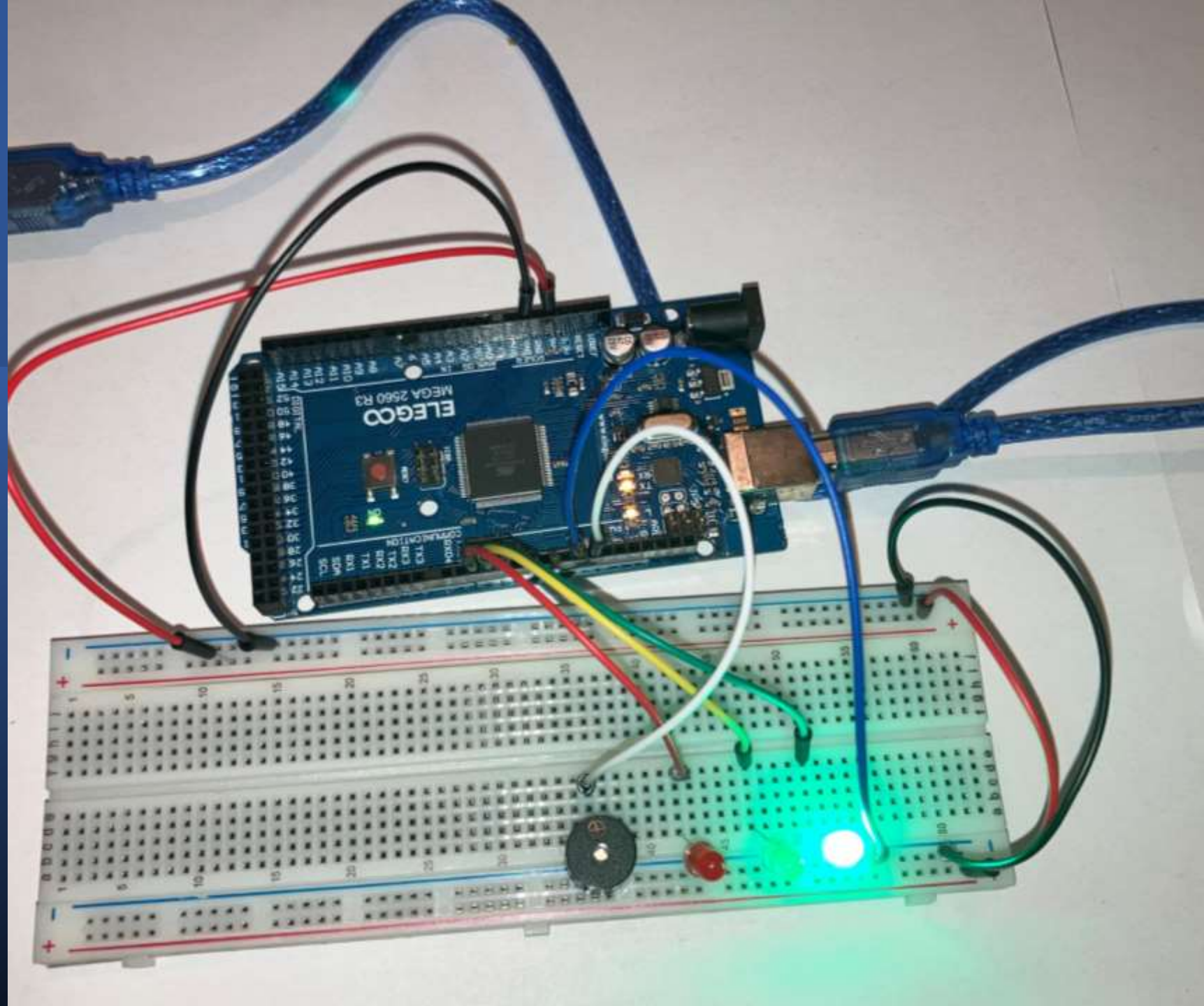
Serial Monitor ×

Message (Enter to send message to 'Arduino Mega or Mega 2560' on 'COM5')

```
CEIS101 Course Project Module 3
Name: Francisco Acosta
Red LED is ON
Red LED is OFF
Red LED is ON
Red LED is OFF
Red LED is ON
Red LED is OFF
Red LED is ON
Red LED is OFF
```

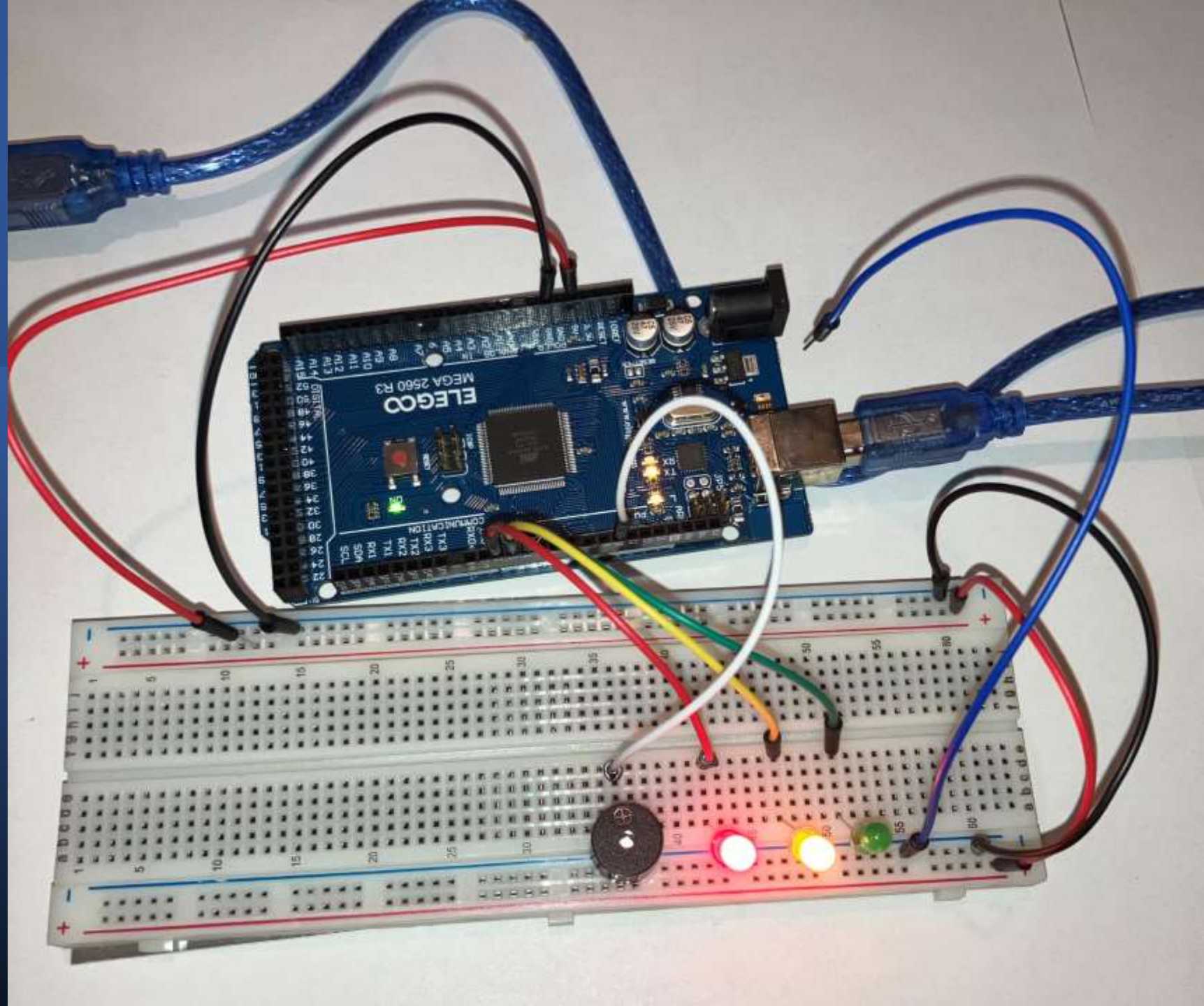
Closed Door

A blue wire connected to the Arduino Mega 2560 was a representation of a closed door.
A green LED light was programmed to illuminate if the door was closed.



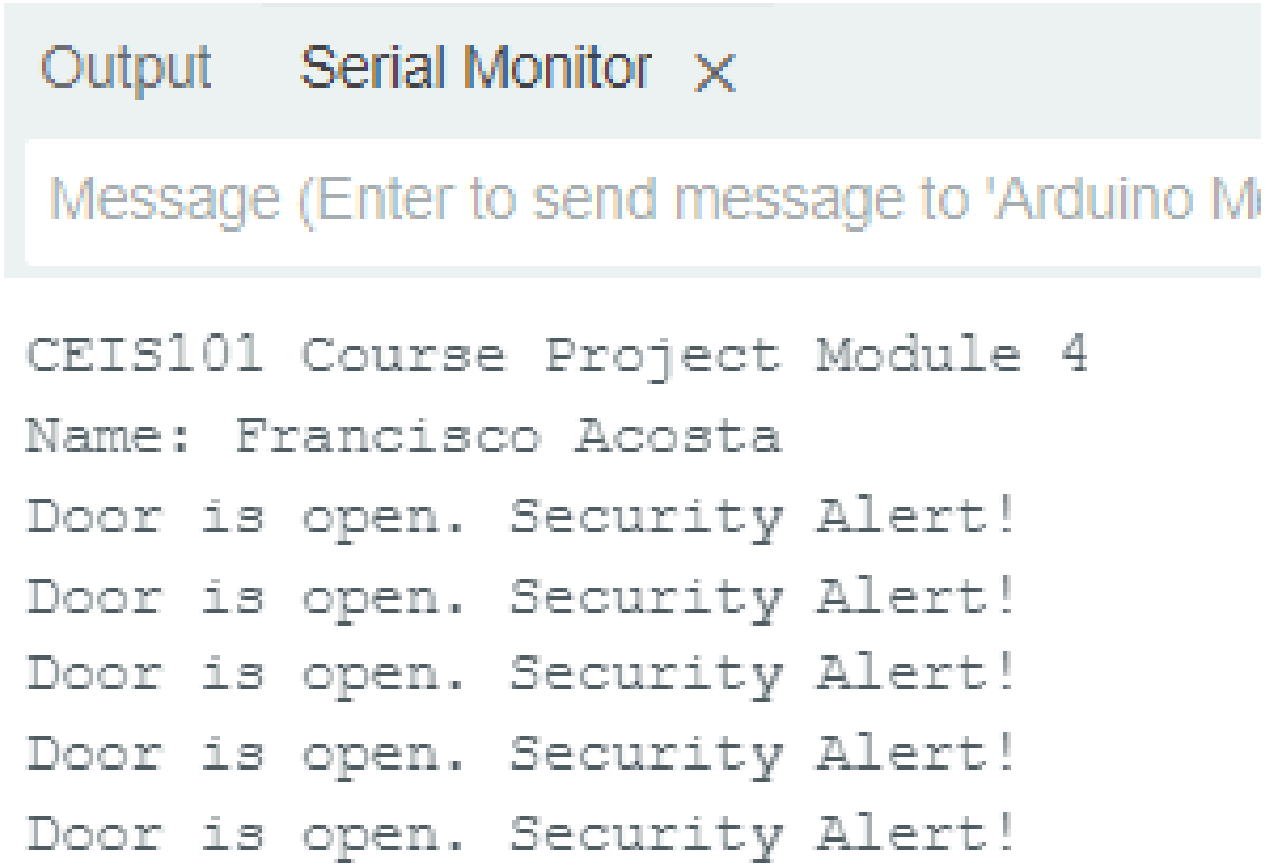
Opened Door

The blue wire disconnected from the Arduino Mega 2560 was a representation of an open door, which would illuminate the yellow and red LEDs while the buzzer radiated an alarming sound.



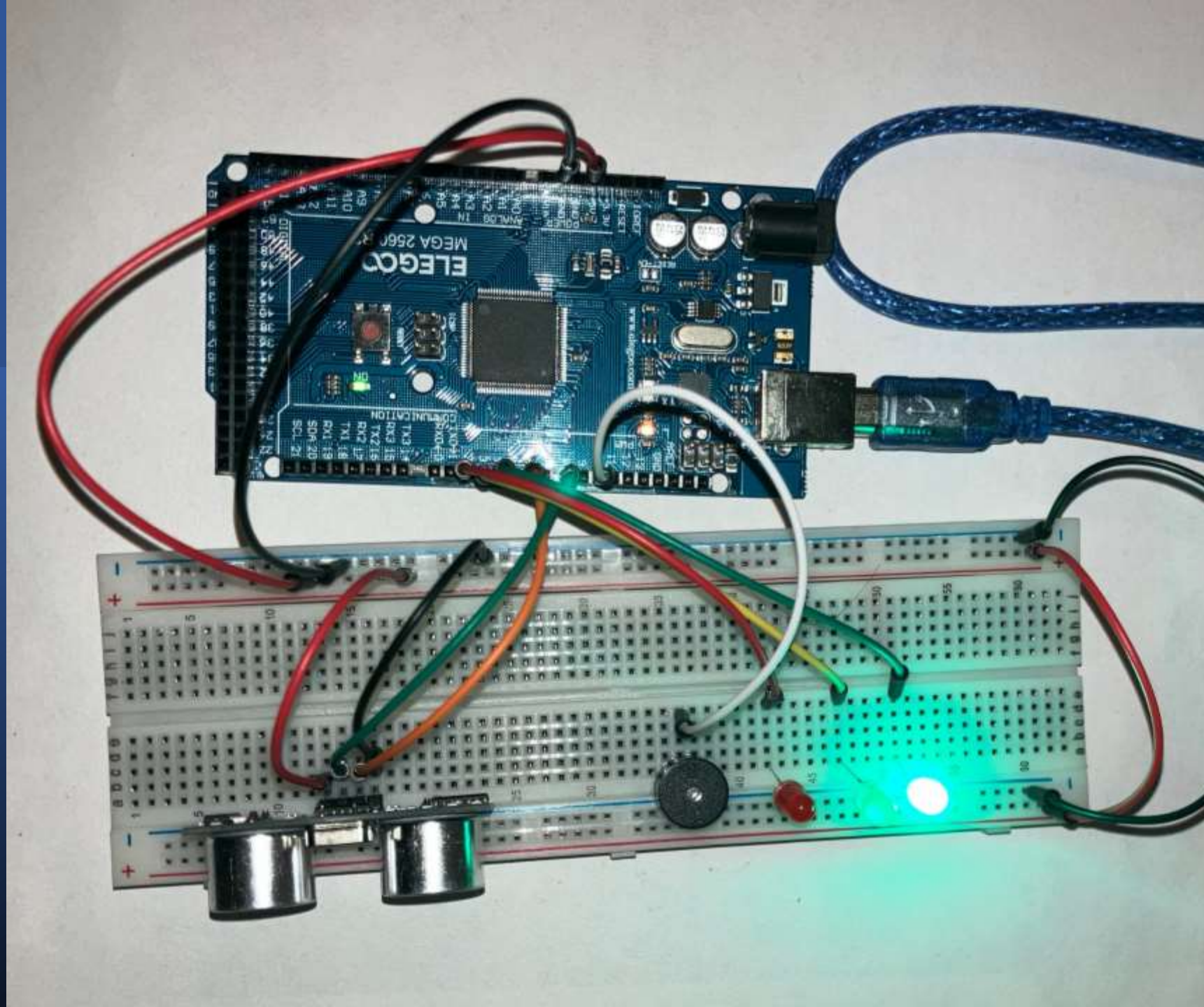
Door
Serial Monitor

Arduino IDE output of the open door.



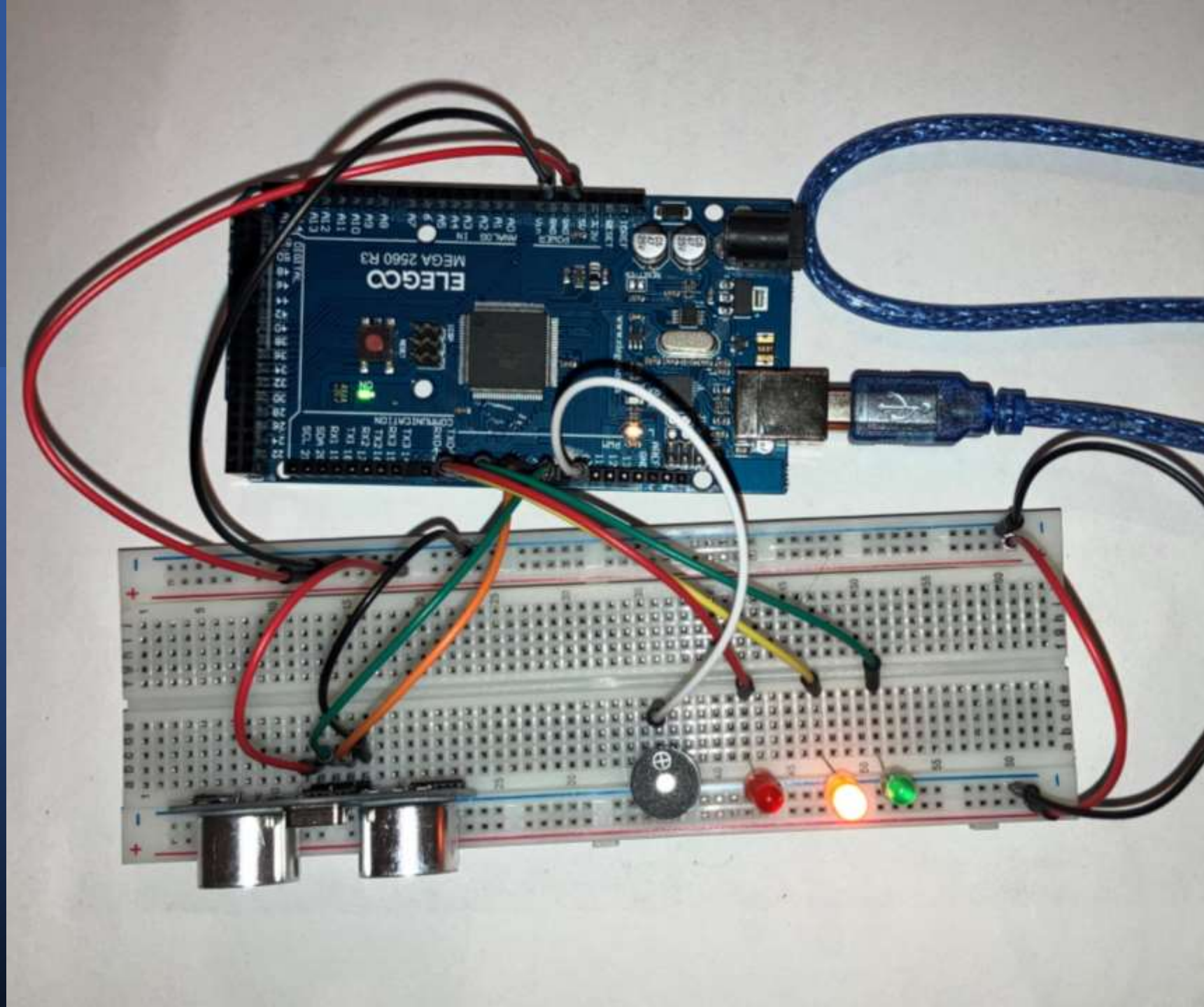
Ultrasonic Sensor

An ultrasonic sensor turned on the green LED light if there was no threat detected within a designated distance.



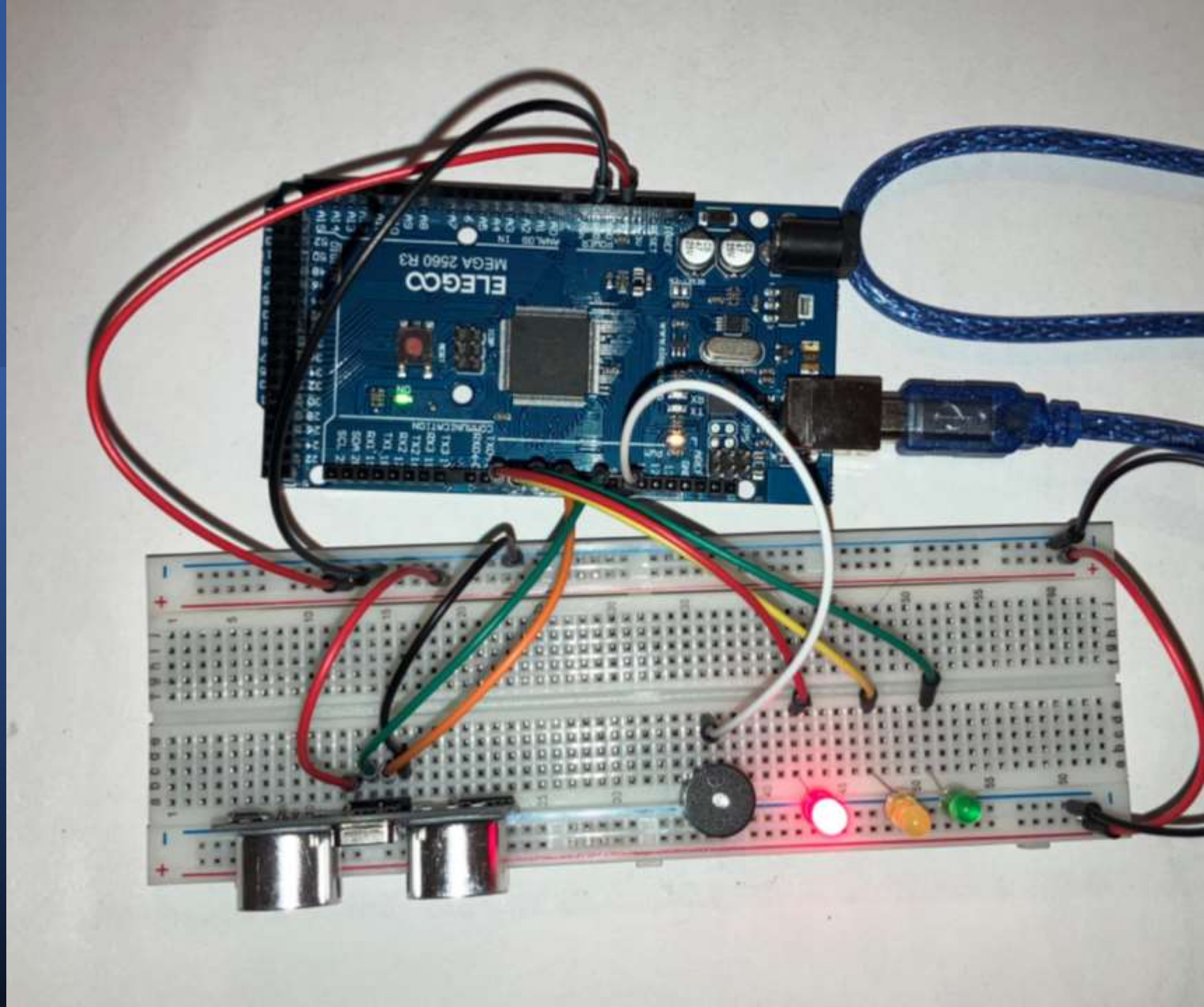
Ultrasonic Sensor

An illuminated yellow LED light denoted a warning of a potential intruder within a particular range.



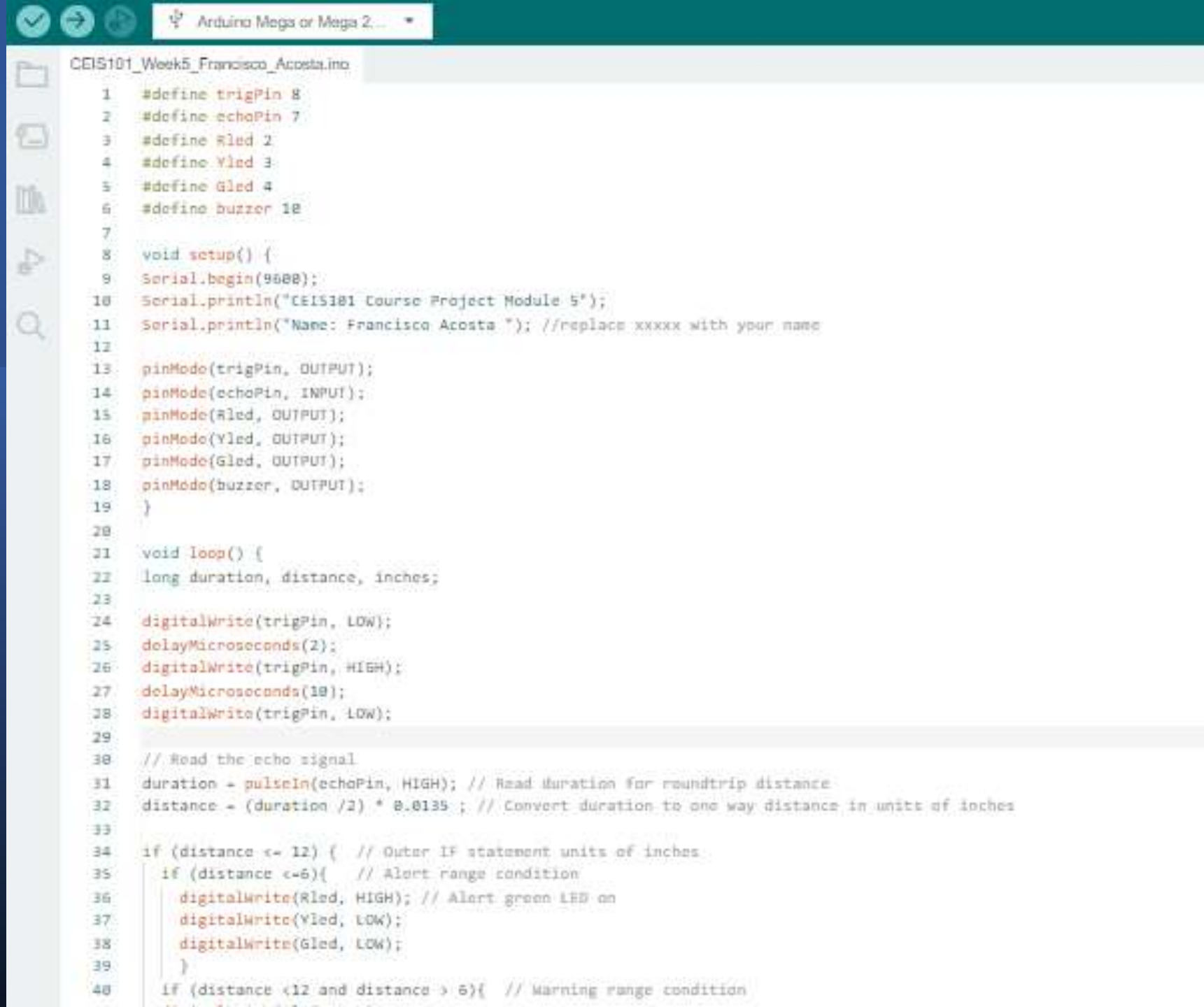
Ultrasonic Sensor

An illuminated red LED light indicated an intruder was indeed present. The buzzer would also radiate a sound.



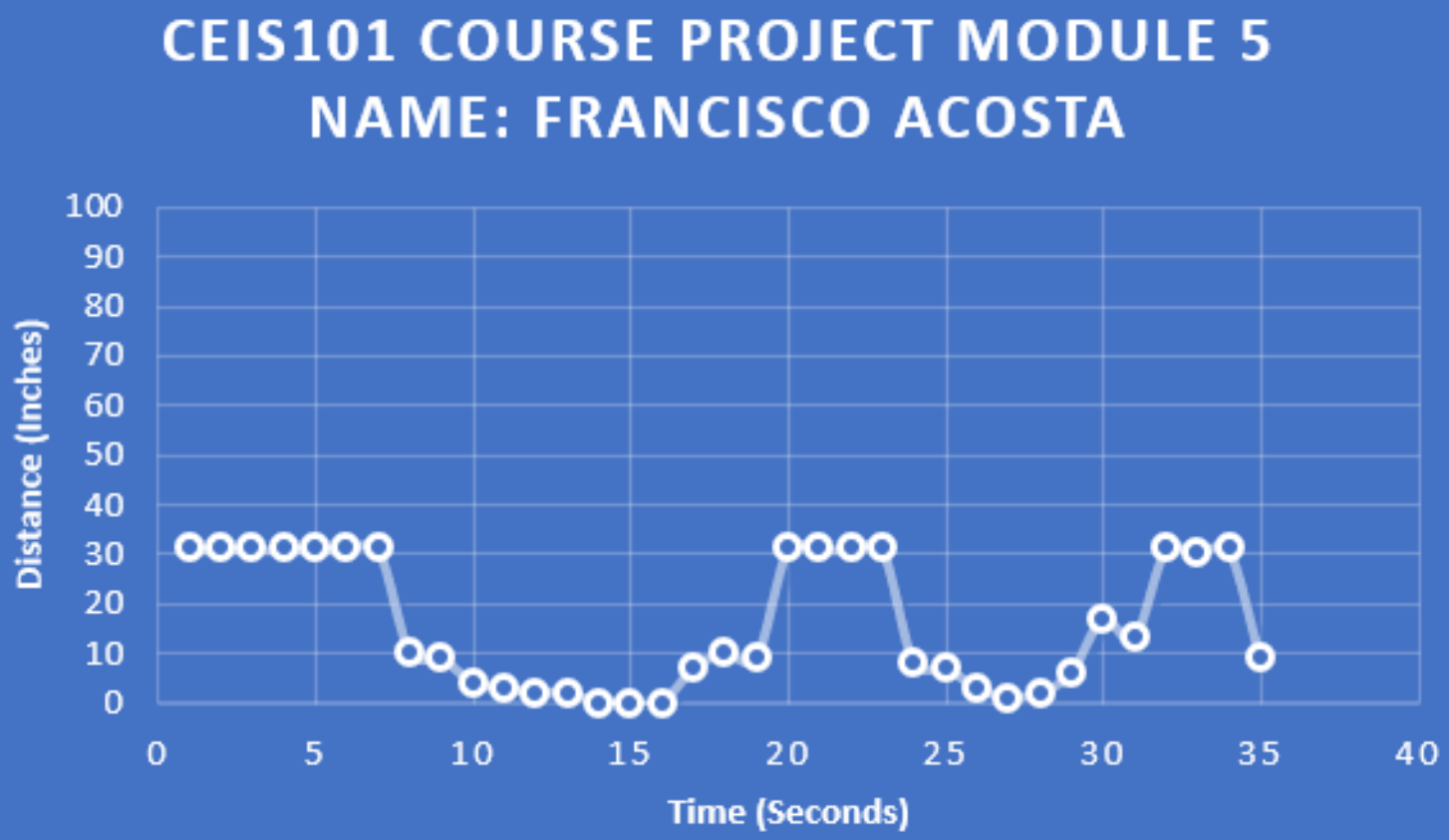
Ultrasonic Sensor Code

This project was developed in the Arduino IDE utilizing the C programming language.



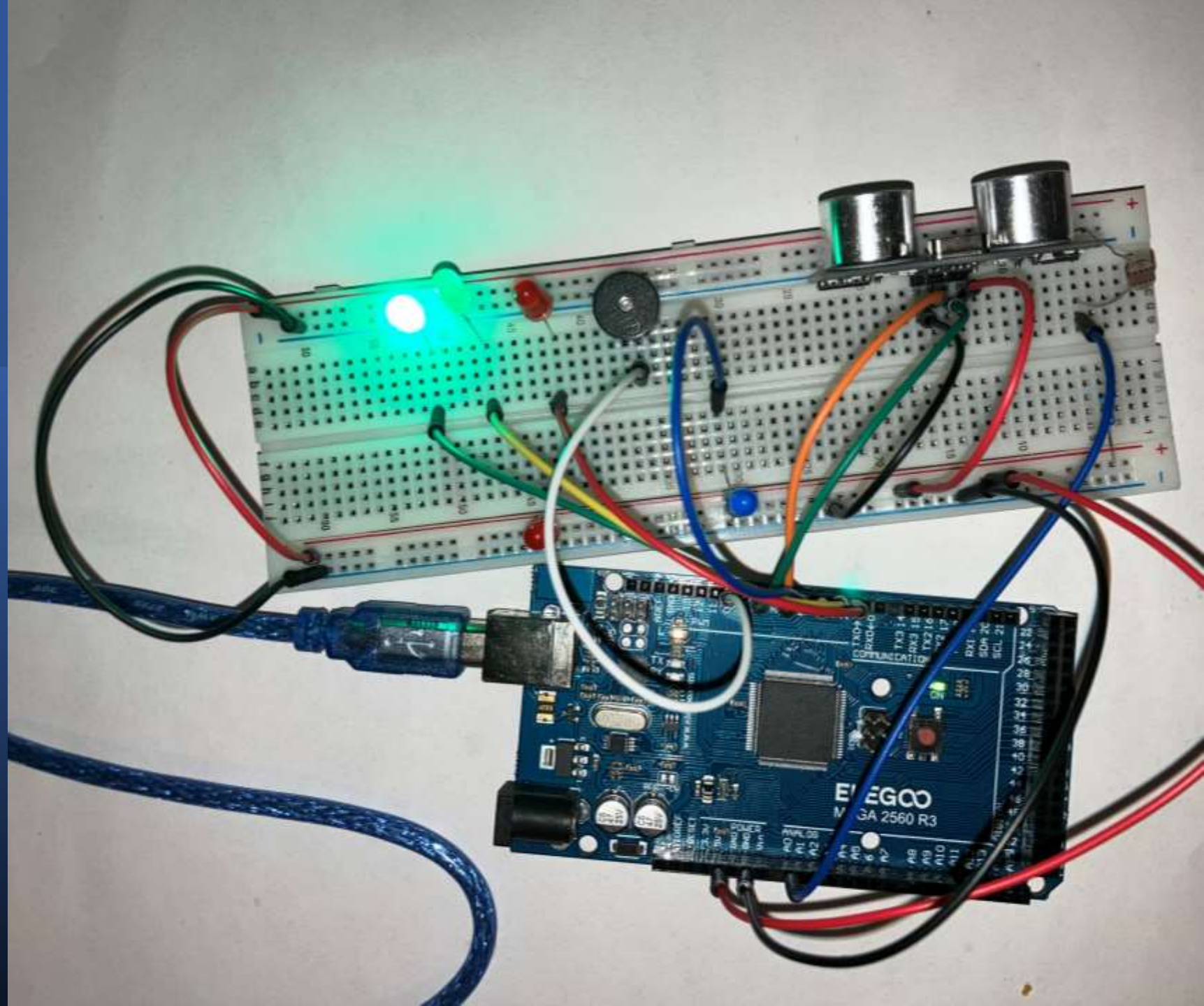
Data Analysis

The ultrasonic sensor produced data indicating the distance from an individual and was converted into a visual representation.



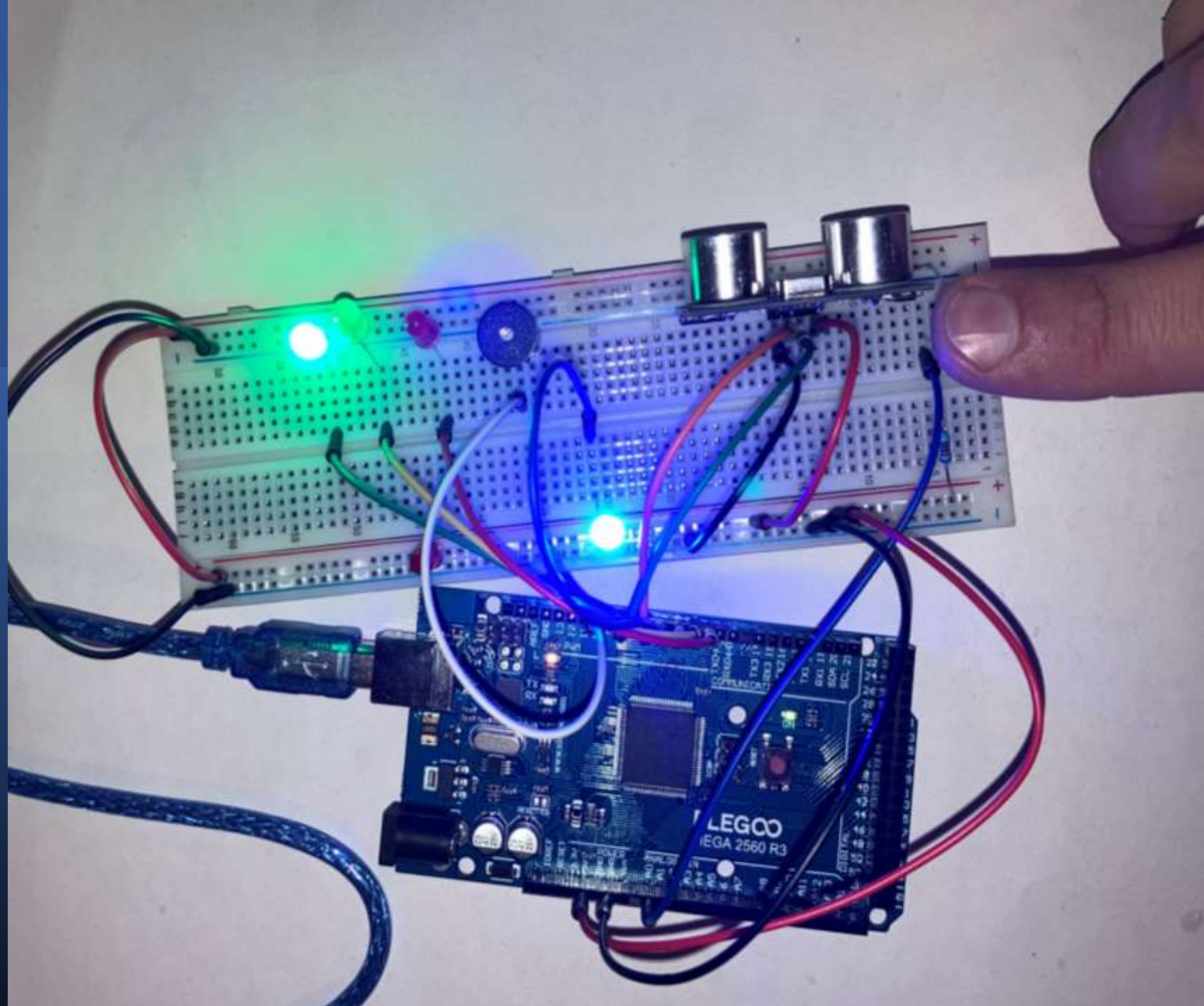
Automated Light

A photoresistor was added to detect any changes in lighting. The green LED light represents its default state with no detection of an individual.



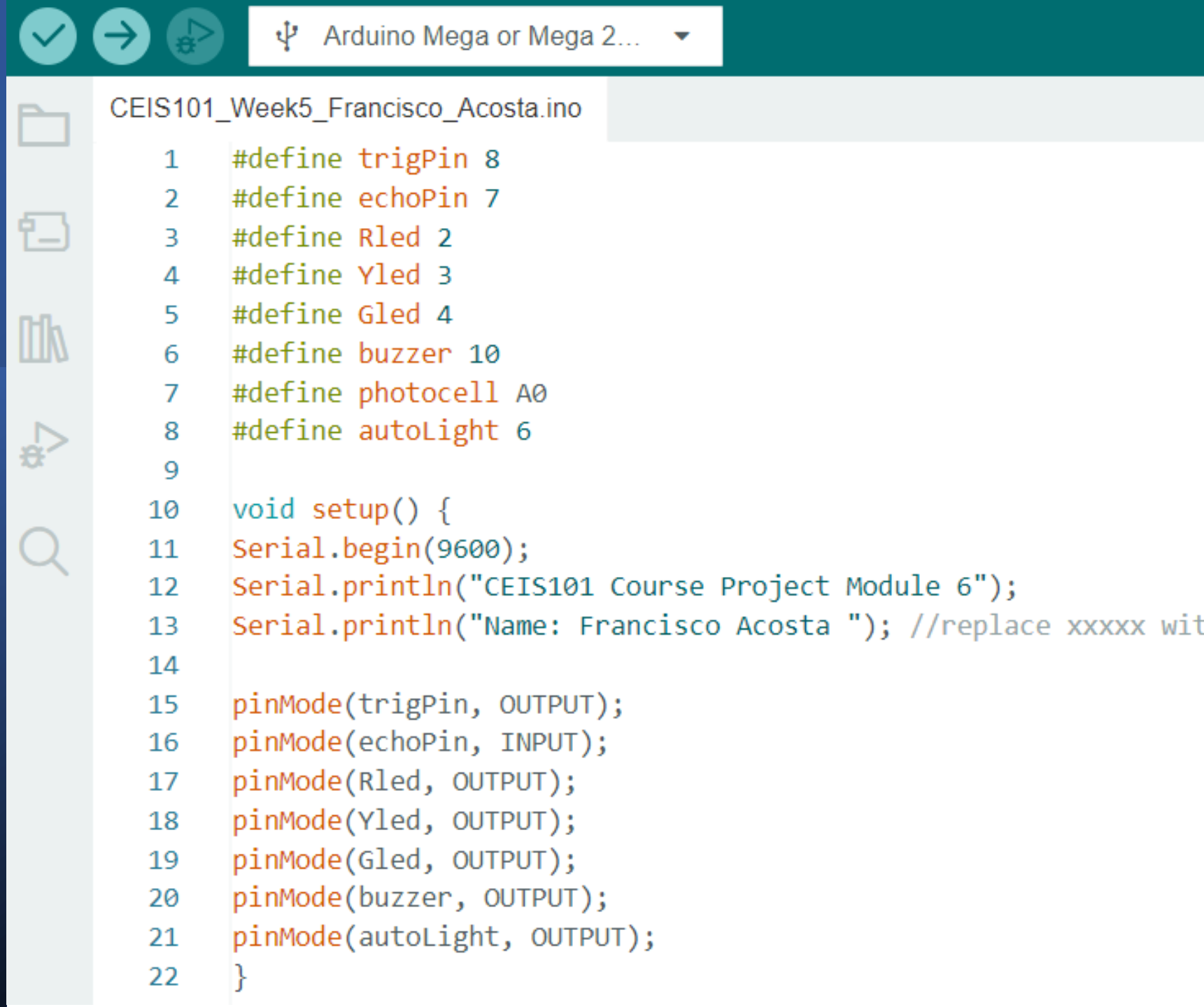
Automated Light

The change in lighting by my finger alerts the blue LED to signify a change in lighting or the detection of an intruder.



Automated Light Code

This project was developed in the Arduino IDE utilizing the C programming language.




```
1   #define trigPin 8
2   #define echoPin 7
3   #define Rled 2
4   #define Yled 3
5   #define Gled 4
6   #define buzzer 10
7   #define photocell A0
8   #define autoLight 6
9
10  void setup() {
11  Serial.begin(9600);
12  Serial.println("CEIS101 Course Project Module 6");
13  Serial.println("Name: Francisco Acosta "); //replace xxxxx wit
14
15  pinMode(trigPin, OUTPUT);
16  pinMode(echoPin, INPUT);
17  pinMode(Rled, OUTPUT);
18  pinMode(Yled, OUTPUT);
19  pinMode(Gled, OUTPUT);
20  pinMode(buzzer, OUTPUT);
21  pinMode(autoLight, OUTPUT);
22  }
```

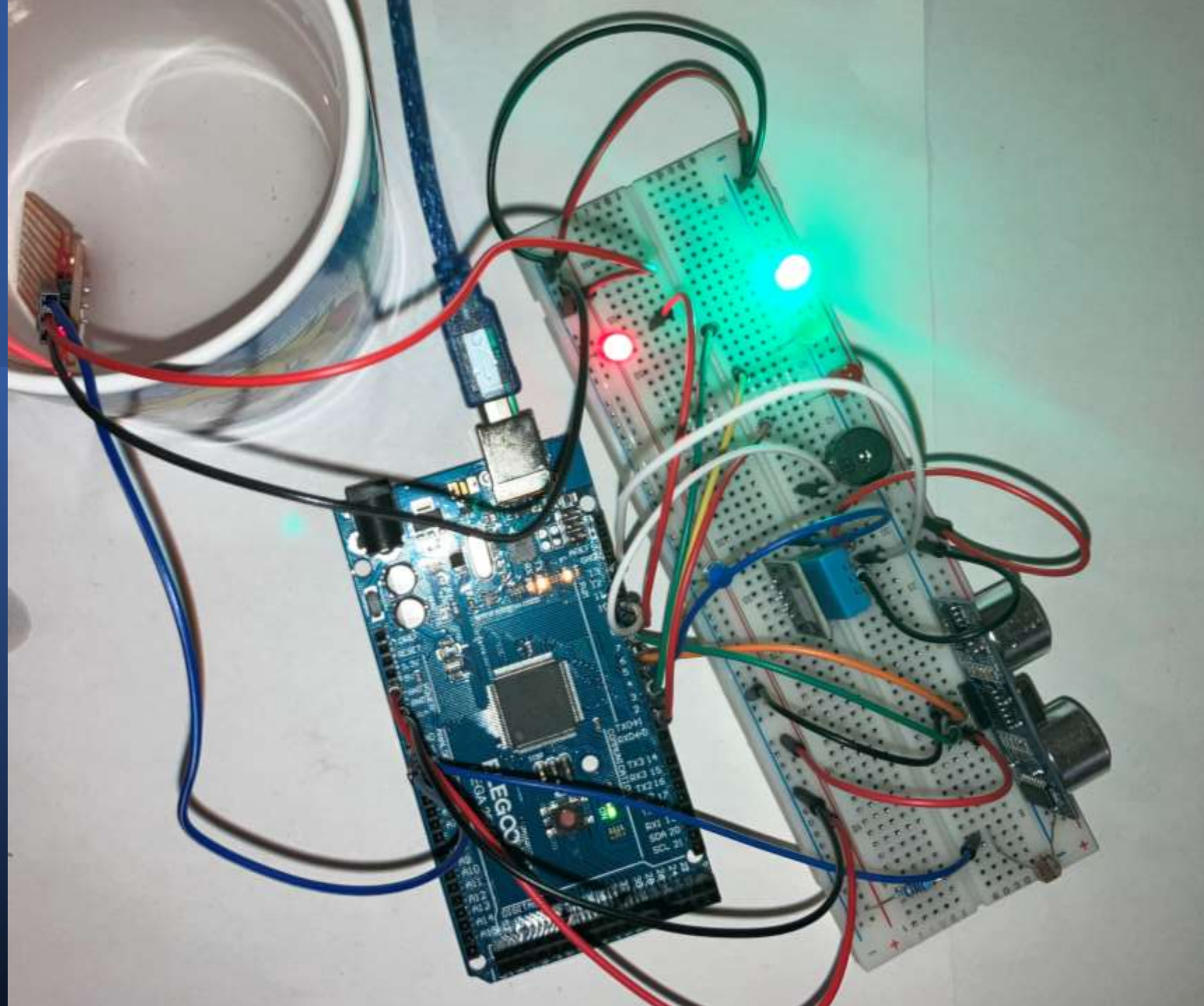
Automated Light Serial Monitor

Information output via the serial monitor in the Arduino IDE.

```
CEIS101 Course Project Module 6
Name: Francisco Acosta
870
The automated light is ON
Alert! Possible Intruder.
864
The automated light is ON
Alert! Possible Intruder.
866
The automated light is ON
Alert! Possible Intruder.
864
The automated light is ON
Alert! Possible Intruder.
```

Water Sensor

A water sensor was added to check for possible flooding which is indicated by one of the red LEDs.



Challenges

One challenge I faced was coming to the discovery the photoresistor needed very particular specifications within the C based code to match the lighting of its surroundings to achieve an accurate reading. Also, finding the appropriate amount of space on the breadboard to include all sensors was a challenging feat.

Overcoming these challenges took patience and persistence. Roadblocks stalled progress, but I was able to persevere to complete the build. Researching documentations was a major contributing factor to my success.

Lessons Learned

I learned to build circuitry with the provided hardware and apply logic using the C programming language to assign specified tasks to the appropriate sensor.

Each sensor was a new journey for me considering it was my first exposure to such devices.

I walked away with the ability to combine hardware and software to complete the task at hand.

Career Skills

Closing the gap between hardware and software was the biggest skill I learned. I believe this will benefit my career immensely due to the versatility it allows me to bring into my skillset. Creating an IoT home security system from scratch enabled me to explore the C based language and earn valuable problem-solving skills to fix future potential problems I may face.

Problem solving itself will open many doors to advance my career considering technology is often viewed as a puzzle and creative solutions are highly valued. The addition of a great work ethic and the ability to efficiently code will aid my endeavors for years to come.

Conclusion

The journey to create an IoT home security system has expanded my skillset and given me valuable knowledge of hardware, programming and data analysis. The ability to complete these tasks illustrate a clear path to the next level and future growth to come.

References

- DeVry documentation
  - Detailed instruction to troubleshoot tasks
- www.tinkercad.com
  - Practice the circuit building process
- www.arduino.cc
  - Documentation to aid the install and understand the IDE
- www.w3schools.com
  - Research to better understand the C programming language